Chemical Kinetics (1/20/17)

1. The decomposition of benzoyl peroxide in diethyl ether, a first-order reaction, is 75.2 percent complete in 10 min at 60 °C.

   a) Derive the expression of integrated rate law for first-order reaction. (5 pts)
   b) Calculate the rate constant for the reaction. (10 pts)
   c) Derive and calculate the half-life of the reaction. (10 pts)

2. The following mechanism (i) for the reaction

$$H_2(g) + I_2(g) \rightarrow 2HI(g)$$

was generally accepted until 1967:

$H_2 + I_2 \rightarrow 2HI$ Rate constant: k

In 1967, it was shown that the reaction probably proceeds by the mechanism (ii):

		Rate constant
$I_2 \rightarrow 2I$	(1)	k_1
$2I \rightarrow I_2$	(2)	k_2
$2I + H_2 \rightarrow 2HI$	(3)	k_3

   a) Obtain the rate law for $d[HI]/dt$ predicted by each of the mechanisms (i) and (ii). Use the steady state approximation where appropriate. (15 pts)
   b) Obtain the rate law predicted by mechanism (ii) under the assumption that reactions (1) and (2) comprise an equilibrium, with K being the equilibrium constant for the reaction $I_2 \leftrightarrow 2I$. Under what conditions do the results of (a) and (b) for mechanism (ii) agree? (15 pts)
   c) By producing I atoms photochemically in known concentration it is possible to measure k_3,

T (K)	417.9	520.1
k_3 ($L^2\ mol^{-2}\ s^{-1}$)	1.12×10^5	4.0×10^5

   (k_3 is defined with reference to $-d[H_2]/dt$.) Calculate the activation energy for reaction (3) in mechanism (ii). (hint: $k = Ae^{-Ea/RT}$) (20 pts)

3. Potassium-argon dating is used in geology and archeology to date sedimentary rocks. Potassium-40 decays by two different paths:

$$^{40}_{19}K \rightarrow ^{40}_{20}Ca + ^{0}_{-1}e \qquad (89.3\% \text{ by mol})$$

$$^{40}_{19}K \rightarrow ^{40}_{18}Ar + ^{0}_{1}e \qquad (10.7\% \text{ by mol})$$

The overall half-life for the delay of potassium-40 is 1.3×10^9 y. Estimate the age of sedimentary rocks with an argon-40 to potassium-40 molar ratio of 0.0102. (25 pts)